

Lab #1

Instructor: Dr. Gökhan Kaya

Assistants: Başak Karakaş, Çağla Nur Yuva, Ümit Murat Akkaya

Note (Ubuntu):

If gcc is not installed, run:

```
sudo apt update
sudo apt install gcc
```

To compile and run a .c file (example: main.c):

```
gcc main.c -o main
./main
```

Allowed Libraries and Functions

- You may use `#include <stdio.h>` and `#include <math.h>`.
- Only `sqrt()` from `math.h` is allowed as built-in function.

Question 1: Expression Evaluation (Input/Output + Basic Arithmetic)

(40 points)

Read **five integers** from the user: a, b, c, d, e .

Compute the following expression and print the final value as a **double**:

$$\text{result} = \frac{a^3 + 2b - cd}{e} + \frac{a + b}{c + 1} - \frac{d}{2}$$

Notes / Hints:

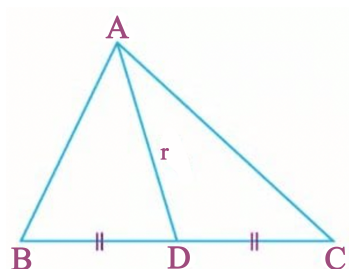
- For a binary operation, the result type is **int** if both operands are integers; otherwise, it is **double**.
- To avoid division by zero, $e \neq 0$ and $c \neq -1$.

Expected program execution:

```
Enter five integers (a, b, c, d, e): 1 2 3 4 5
Integers: a=1, b=2, c=3, d=4, e=5
Final value: -2.6500
```

Question 2: Points in 2D (Midpoint, Distance, Circle Circumference)

(60 points)



Consider the triangle with vertices A , B , and C as shown in the figure. Point D is the **midpoint** of segment BC (i.e., $BD = DC$). The segment AD is labeled as r .

The coordinates are given as

$$A(x_1, y_1), \quad B(x_2, y_2), \quad C(x_3, y_3).$$

(a) Define the coordinates as the following integers: $x_1 = 2$, $y_1 = 8$, $x_2 = -4$, $y_2 = 0$, $x_3 = 10$, $y_3 = 0$.

(b) Compute the coordinates of $D(x_D, y_D)$ as double using:

$$x_D = \frac{x_2 + x_3}{2.0}, \quad y_D = \frac{y_2 + y_3}{2.0}.$$

(c) Using the following distance equation compute the value of r :

$$r = \sqrt{(x_1 - x_D)^2 + (y_1 - y_D)^2}.$$

(d) Compute the circumference of a circle of radius r .

$$C = 2\pi r.$$

Important: Define π as a macro and use the value 3.14.

Expected program execution:

```
The Coordinates: A(2,8) B(-4,0) C(10,0)

Coordinates of Midpoint D: (3.000, 0.000)
Length of segment AD (r): 8.062
Circumference (C): 50.631
```